Federated Self-training for Semi-supervised Audio Recognition

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01 Context & approach

The challenge

Most audio collected on devices has no labels. *FedSTAR* lets this unused audio contribute to a shared model by combining labeled examples with confident local predictions used as temporary labels.

Our contribution

Learn jointly from labeled and unlabeled audio on each client.
Filter pseudo-labels with temperature scaling and an increasing confidence threshold.
Test label budgets, client distributions and optional public-data pretraining.

1 Share the current model

Broadcast one global model to clients; optional public-audio pretraining supplies its initial weights.

2 Combine two learning signals

The same local model predicts pseudo-labels, filters uncertain examples, and combines supervised and weighted pseudo-label losses in one backward pass.

3 Average and repeat

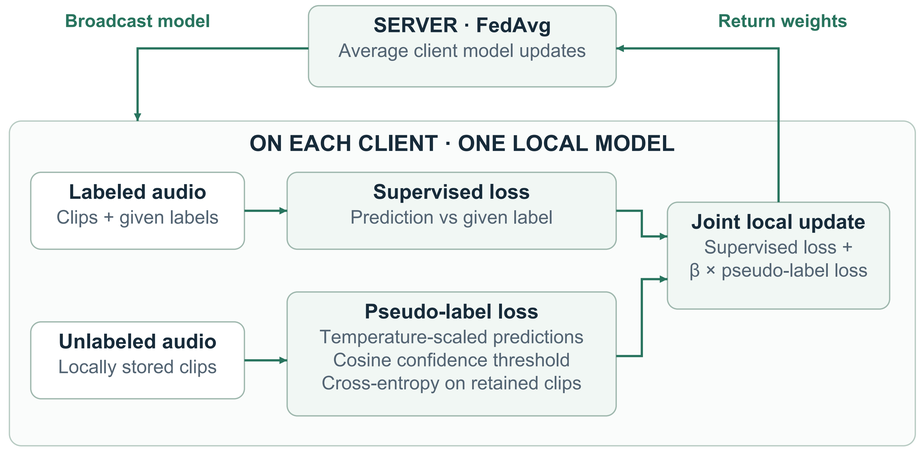
Use FedAvg to combine model updates and begin the next round; raw downstream audio stays local.

Scope

Gains vary by task and label budget; public-data pretraining does not always improve final accuracy. Client training is simulated; class-skew tests retain randomly distributed unlabeled clips.

02 Method

Use confident pseudo-labels to learn from audio that remains unlabeled on devices.



Compute both losses before the same backward update.

Repeat across selected clients and federated rounds. Raw downstream audio stays local.

Core formulation

$$\mathcal{L}_k = \mathcal{L}_{\mathrm{CE}}(y, p_{\theta^k}(x_l)) + \beta \mathcal{L}_{\mathrm{CE}}(\hat{y}, p_{\theta^k}(x_u))$$

One client model uses true labels y and confident pseudo-labels ŷ. β weights the unlabeled-audio loss; both terms share one backward pass.

Paper

Code

03 Results

3%
labeled data
15-client comparison

86.82%
Speech Commands accuracy
FedSTAR at 3% labels

Learning with 3% labels

Dataset	Supervised FL (%)	FedSTAR (%) (Ours)
Ambient Context	33.03 ± 0.002	**49.54 ± 0.005**
Speech Commands	62.98 ± 0.003	**86.82 ± 0.006**
VoxForge	54.26 ± 0.002	**55.82 ± 0.011**

3% labels, 15 clients, 100 rounds, 80% participation. Three-run mean accuracy; printed ± terms are described as variance.

Pretraining across label budgets

Labels	Random init. (%)	Pretrained (%)
3%	85.40 ± 0.008	85.17 ± 0.001
5%	88.52 ± 0.005	88.28 ± 0.002
20%	92.24 ± 0.012	93.67 ± 0.005
50%	92.26 ± 0.005	93.83 ± 0.003

Speech Commands: 50% of remaining unlabeled clips, 15 clients, 100 rounds. Three-run means; ± terms denote reported variance.

Takeaway

At 3% labels, **self-training improves all three tasks**, but the gains vary widely. Public-data pretraining can help early convergence, while its **final-accuracy benefit depends on the data budget**.